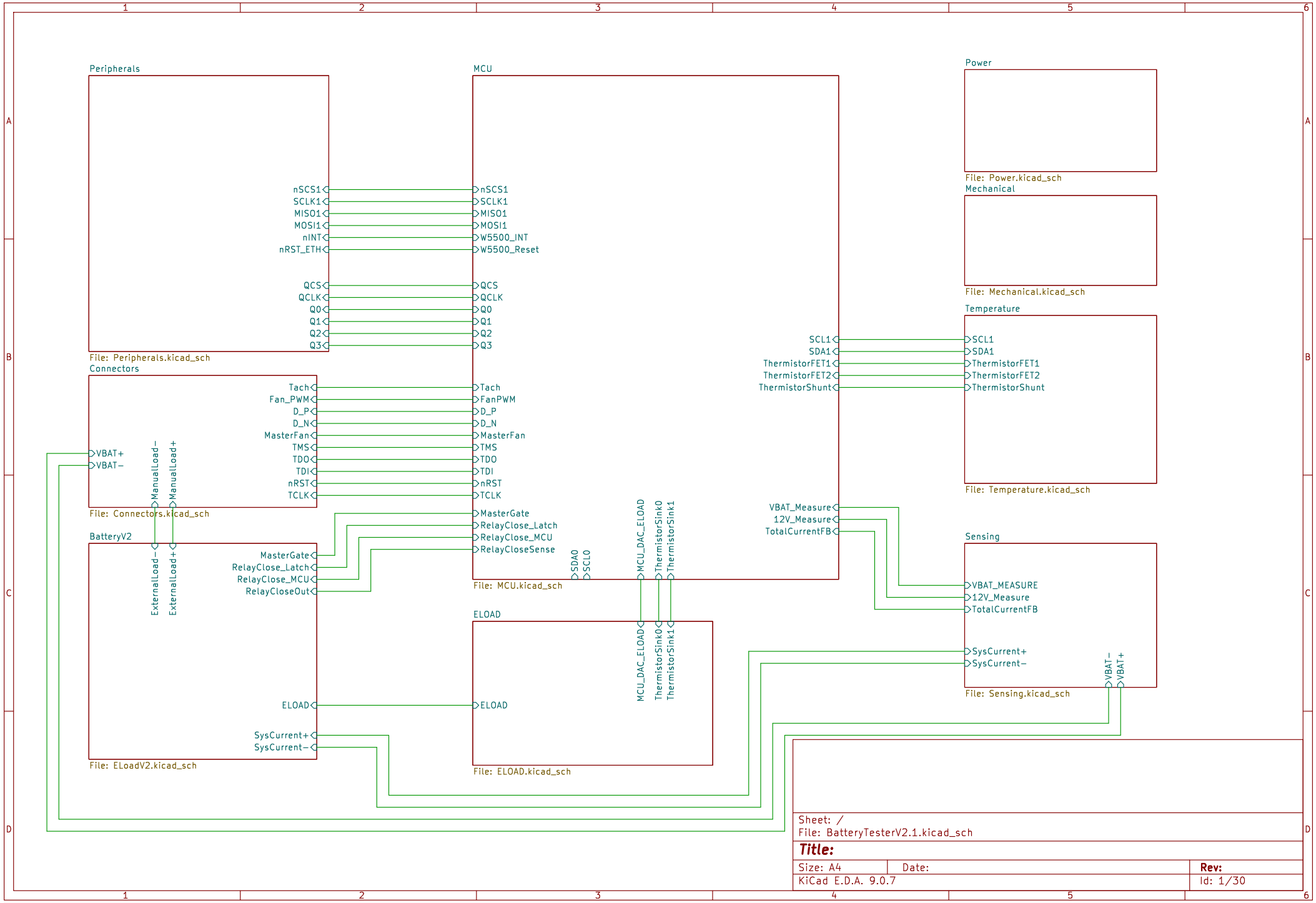
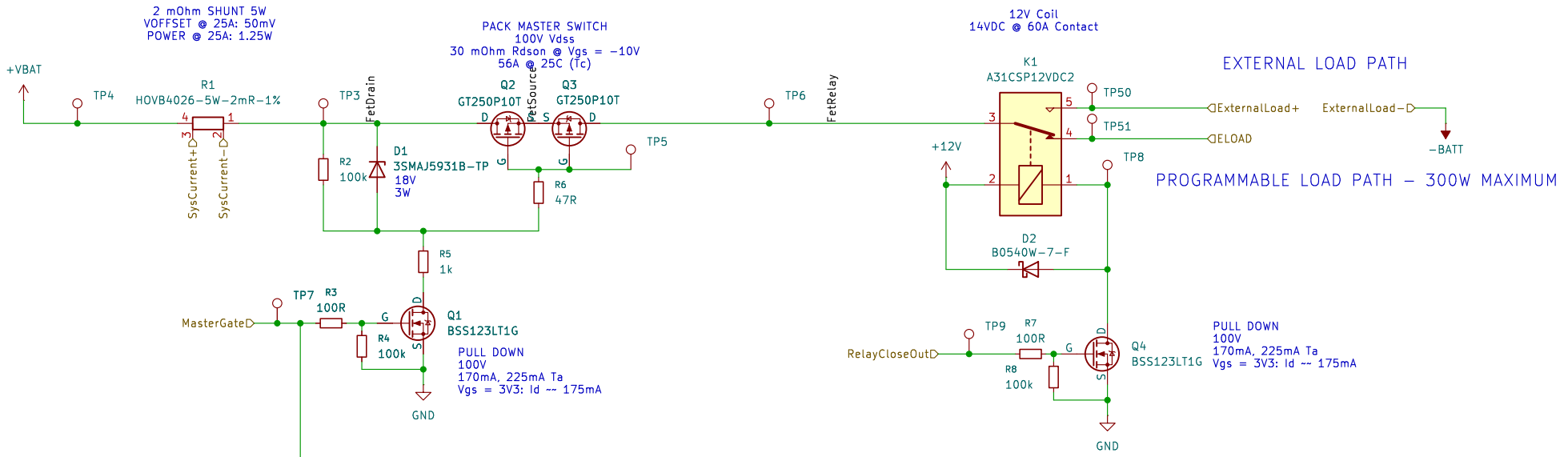
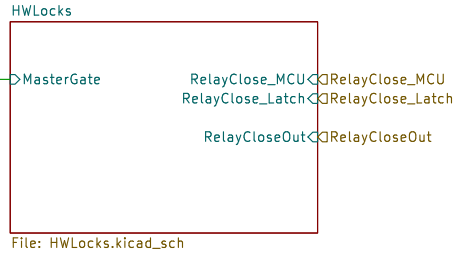

Peripherals
nSCS1
SCLK1
MISO1
MOSI1
nINT
nRST_ETH
QCS
QCLK
Q0
Q1
Q2
Q3
File: Peripherals.kicad_sch
MCU
nSCS1
SCLK1
MISO1
MOSI1
W5500_INT
W5500_Reset
QCS
QCLK
Q0
Q1
Q2
Q3
Tach
FanPWM
D_P
D_N
MasterFan
TMS
TDO
TDI
nRST
TCLK
MasterGate
RelayClose_Latch
RelayClose_MCU
RelayCloseSense
SDA0
SCL0
MCU_DAC_ELOAD
ThermistorSink0
ThermistorSink1
SCL1
SDA1
ThermistorFET1
ThermistorFET2
ThermistorShunt
VBAT_Measure
12V_Measure
TotalCurrentFB
File: MCU.kicad_sch
Power
File: Power.kicad_sch
Mechanical
File: Mechanical.kicad_sch
Temperature
SCL1
SDA1
ThermistorFET1
ThermistorFET2
ThermistorShunt
File: Temperature.kicad_sch
Connectors
VBAT+
VBAT−
ManualLoad−
ManualLoad+
Tach
Fan_PWM
D_P
D_N
MasterFan
TMS
TDO
TDI
nRST
TCLK
File: Connectors.kicad_sch
BatteryV2
ExternalLoad−
ExternalLoad+
MasterGate
RelayClose_Latch
RelayClose_MCU
RelayCloseOut
ELOAD
SysCurrent+
SysCurrent−
File: ELoadV2.kicad_sch
ELOAD
MCU_DAC_ELOAD
ThermistorSink0
ThermistorSink1
ELOAD
File: ELOAD.kicad_sch
Sensing
VBAT_MEASURE
12V_Measure
TotalCurrentFB
SysCurrent+
SysCurrent−
VBAT−
VBAT+
File: Sensing.kicad_sch
Sheet: /
File: BatteryTesterV2.1.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 1/30

BATTERY DISCHARGE: 1S–12S, 3V – 51V, 25A MAXIMUM

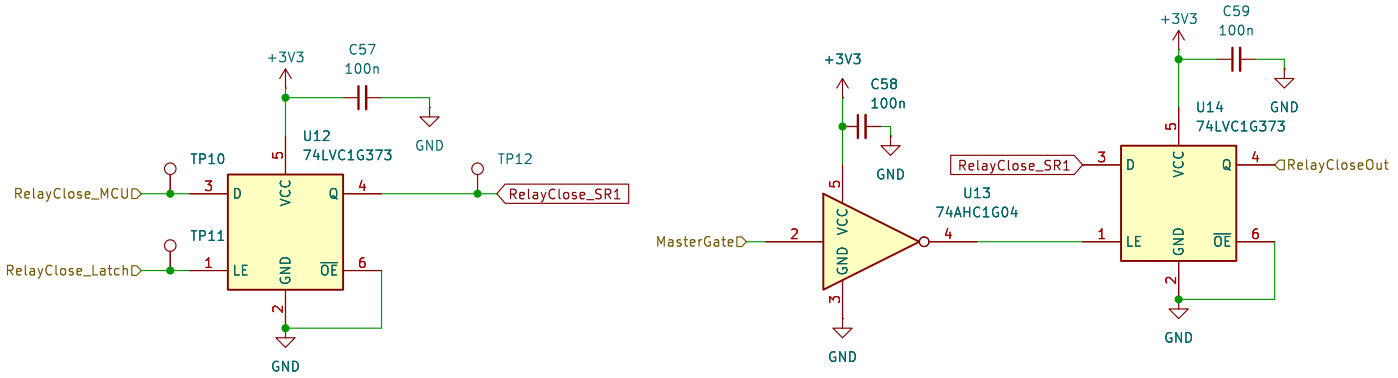


HARDWARE SAFETY SWITCHING

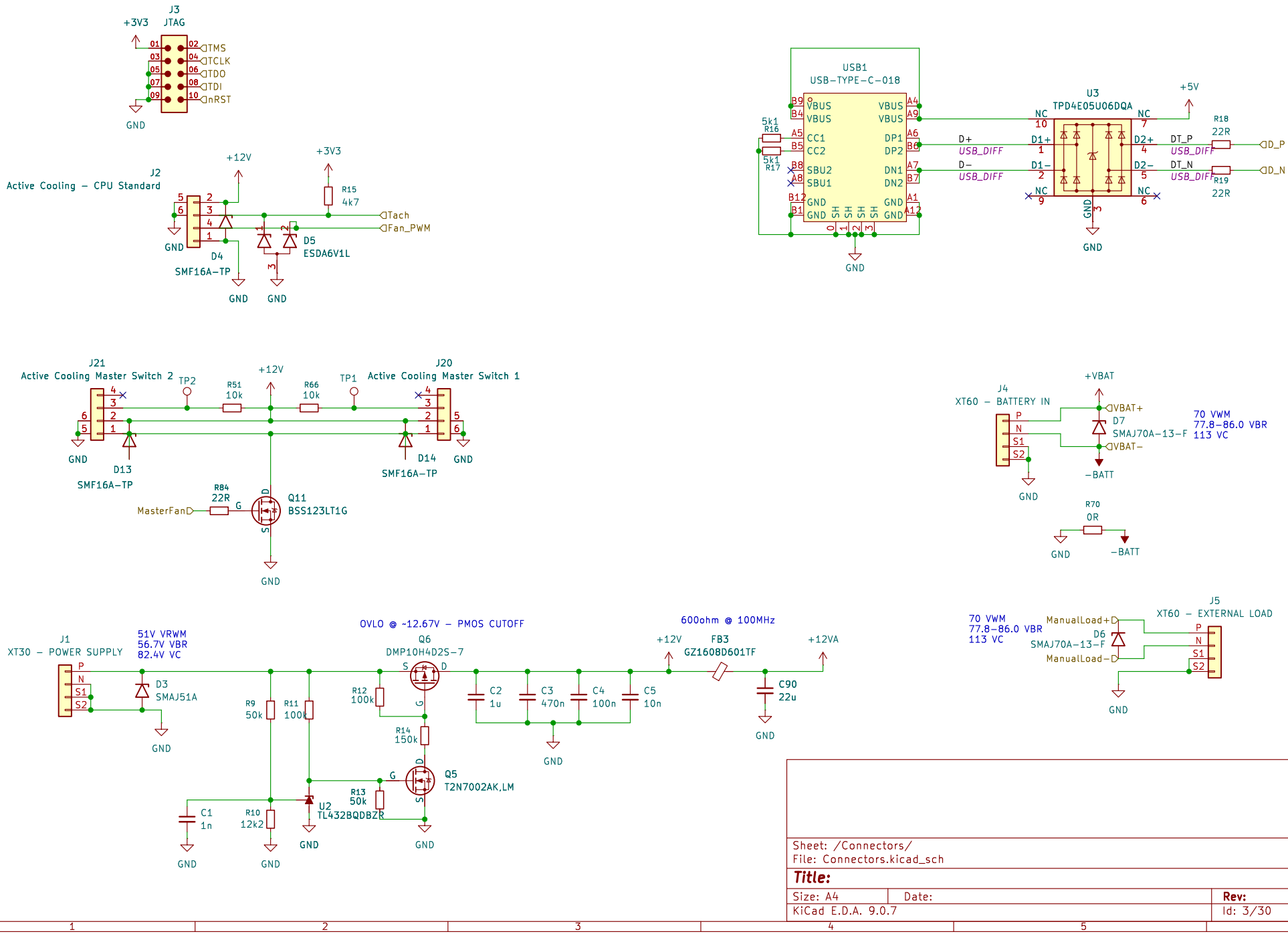


Overcurrent protection:
30A soft overcurrent (FW)
40A hard overcurrent (analog): < 1.5us

Sheet: /BatteryV2/ File: ELoadV2.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 2/30



Sheet: /BatteryV2/HWLocks/		
File: HWLocks.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 10/30


J3
JTAG
+3V3
TMS
TCLK
TDO
TDI
nRST
GND
J2
Active Cooling – CPU Standard
+12V
+3V3
R15
4k7
Tach
Fan_PWM
D4
SMF16A-TP
D5
ESDA6V1L
GND
USB1
USB-TYPE-C-018
5k1
R16
R17
D+
D–
USB_DIFF
U3
TPD4E05U06DQA
+5V
R18
22R
R19
DT_P
DT_N
D_P
D_N
J21
Active Cooling Master Switch 2
J20
Active Cooling Master Switch 1
TP1
TP2
R51
10k
R66
D13
D14
R84
22R
MasterFan
Q11
BSS123LT1G
J4
XT60 – BATTERY IN
+VBAT
VBAT+
VBAT–
D7
SMAJ70A-13-F
70 VWM
77.8-86.0 VBR
113 VC
–BATT
R70
0R
J1
XT30 – POWER SUPPLY
51V VRWM
56.7V VBR
82.4V VC
D3
SMAJ51A
OVLO @ ~12.67V – PMOS CUTOFF
Q6
DMP10H4D2S-7
600ohm @ 100MHz
FB3
GZ1608D601TF
+12V
+12VA
C2
1u
C3
470n
C4
100n
C5
10n
C90
22u
R9
50k
R11
100k
R12
100k
R14
150k
R13
50k
Q5
T2N7002AK,LM
U2
TL432BQDBZR
C1
1n
R10
12k2
J5
XT60 – EXTERNAL LOAD
ManualLoad+
ManualLoad–
D6
SMAJ70A-13-F
Sheet: /Connectors/
File: Connectors.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
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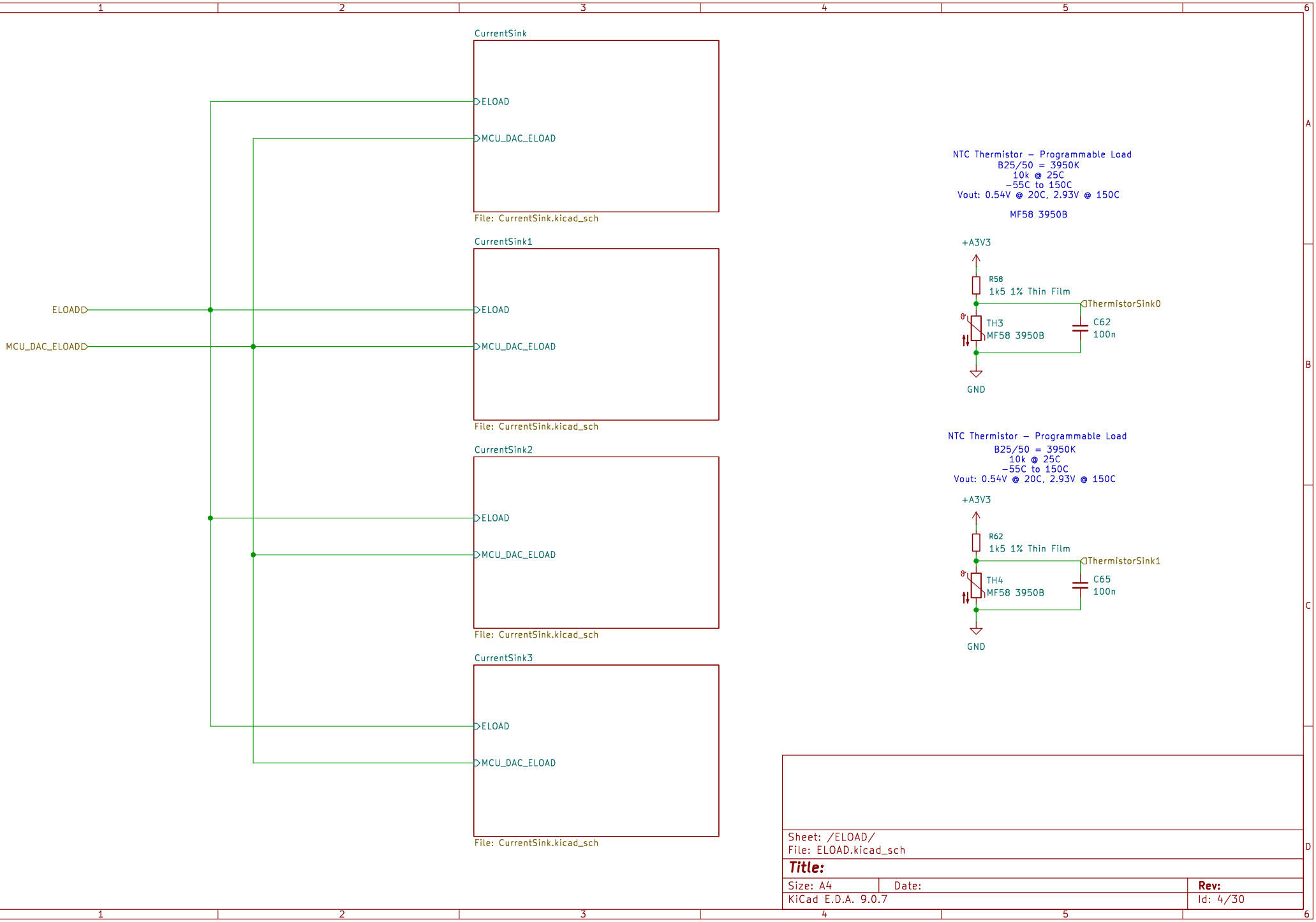

CurrentSink
ELOAD
MCU_DAC_ELOAD
File: CurrentSink.kicad_sch
CurrentSink1
ELOAD
MCU_DAC_ELOAD
File: CurrentSink.kicad_sch
CurrentSink2
ELOAD
MCU_DAC_ELOAD
File: CurrentSink.kicad_sch
CurrentSink3
ELOAD
MCU_DAC_ELOAD
File: CurrentSink.kicad_sch
ELOAD
MCU_DAC_ELOAD
NTC Thermistor – Programmable Load
B25/50 = 3950K
10k @ 25C
−55C to 150C
Vout: 0.54V @ 20C, 2.93V @ 150C
MF58 3950B
+A3V3
R58
1k5 1% Thin Film
ThermistorSink0
TH3
MF58 3950B
C62
100n
GND
NTC Thermistor – Programmable Load
B25/50 = 3950K
10k @ 25C
−55C to 150C
Vout: 0.54V @ 20C, 2.93V @ 150C
+A3V3
R62
1k5 1% Thin Film
ThermistorSink1
TH4
MF58 3950B
C65
100n
GND
Sheet: /ELOAD/
File: ELOAD.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 4/30

Figure 1. On-Region Characteristics

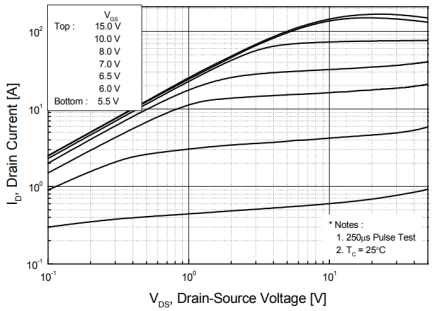


Figure 9. Maximum Safe Operating Area

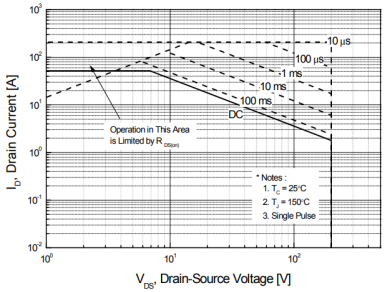
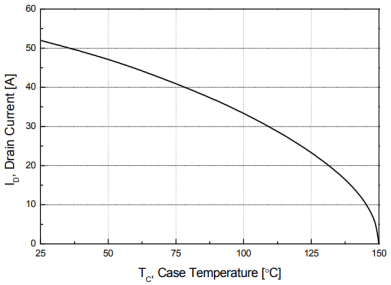
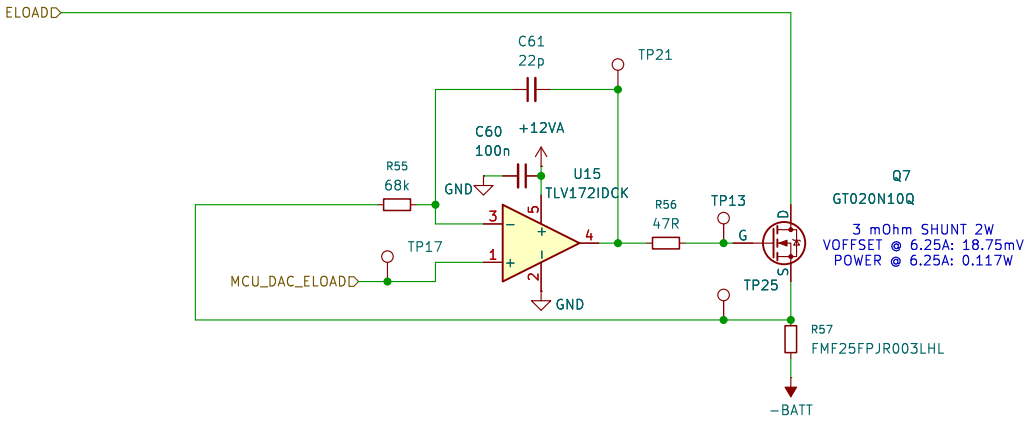


Figure 10. Maximum Drain Current vs. Case Temperature





Sheet: /ELOAD/CurrentSink/
File: CurrentSink.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 11/30

Figure 1. On-Region Characteristics

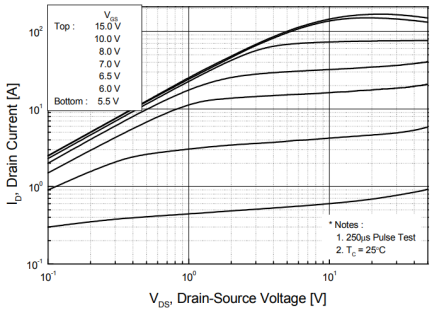


Figure 9. Maximum Safe Operating Area

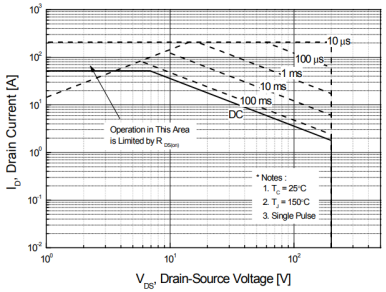
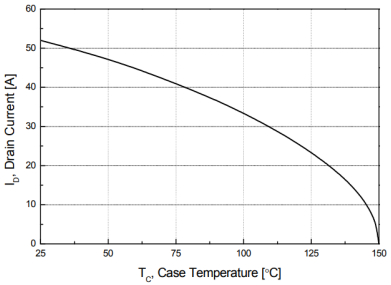
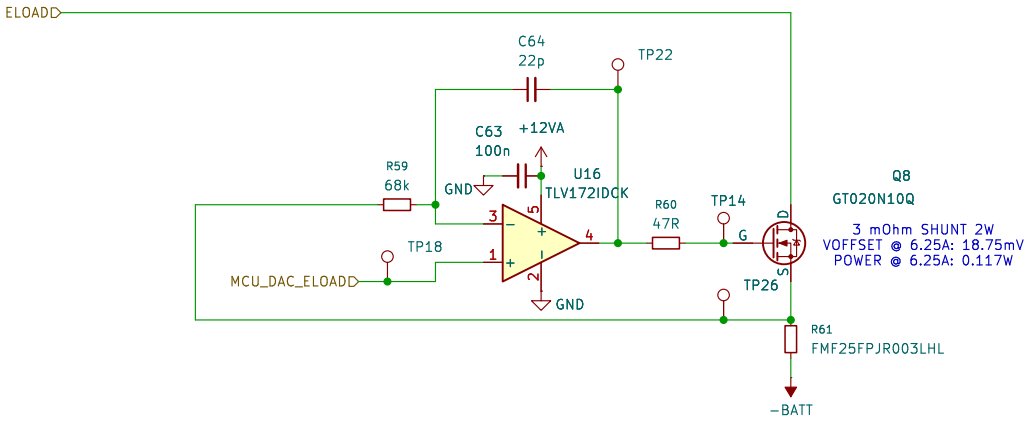


Figure 10. Maximum Drain Current vs. Case Temperature





Sheet: /ELOAD/CurrentSink1/
File: CurrentSink.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 12/30

Figure 1. On-Region Characteristics

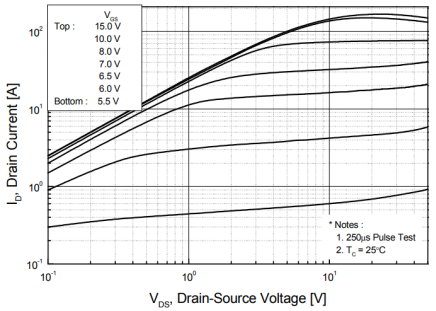


Figure 9. Maximum Safe Operating Area

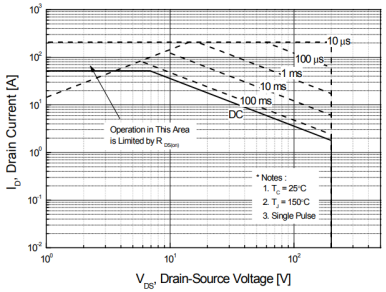
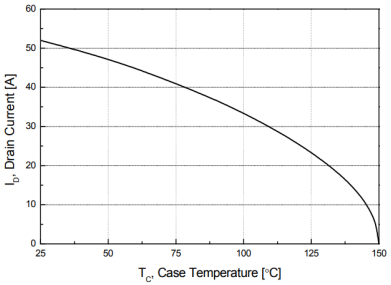
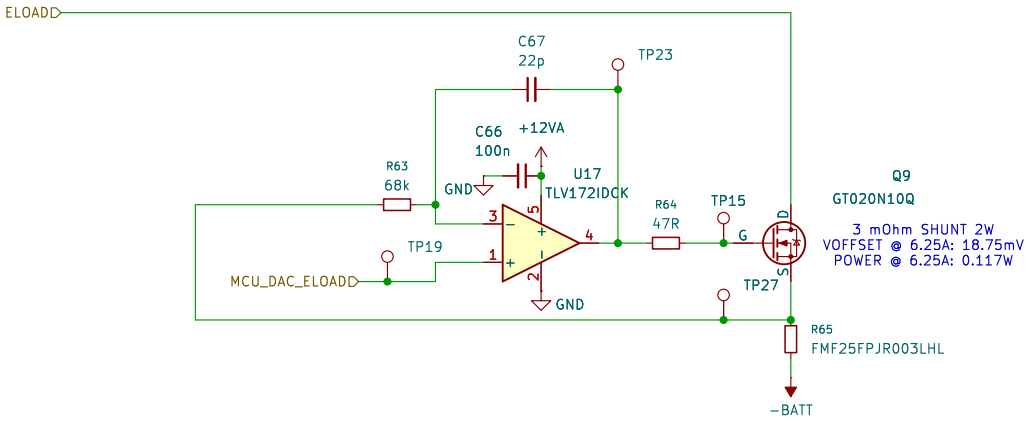


Figure 10. Maximum Drain Current vs. Case Temperature





Sheet: /ELOAD/CurrentSink2/
File: CurrentSink.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 13/30

Figure 1. On-Region Characteristics

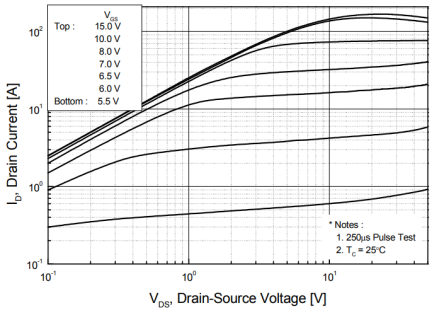


Figure 9. Maximum Safe Operating Area

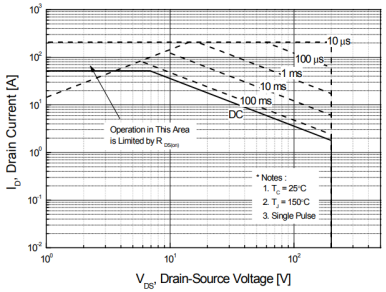
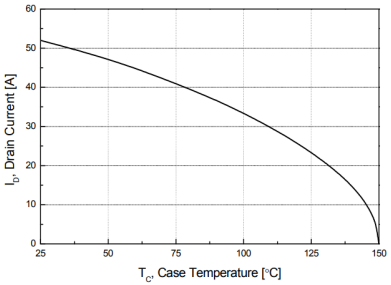
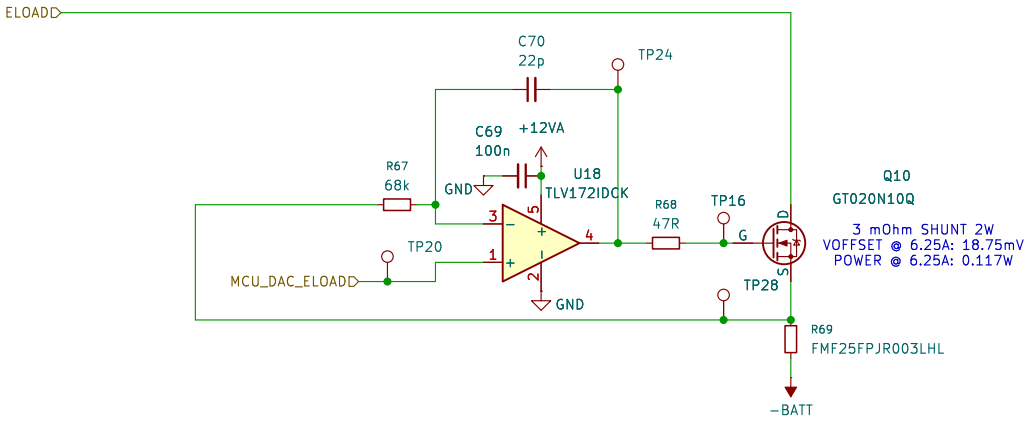


Figure 10. Maximum Drain Current vs. Case Temperature

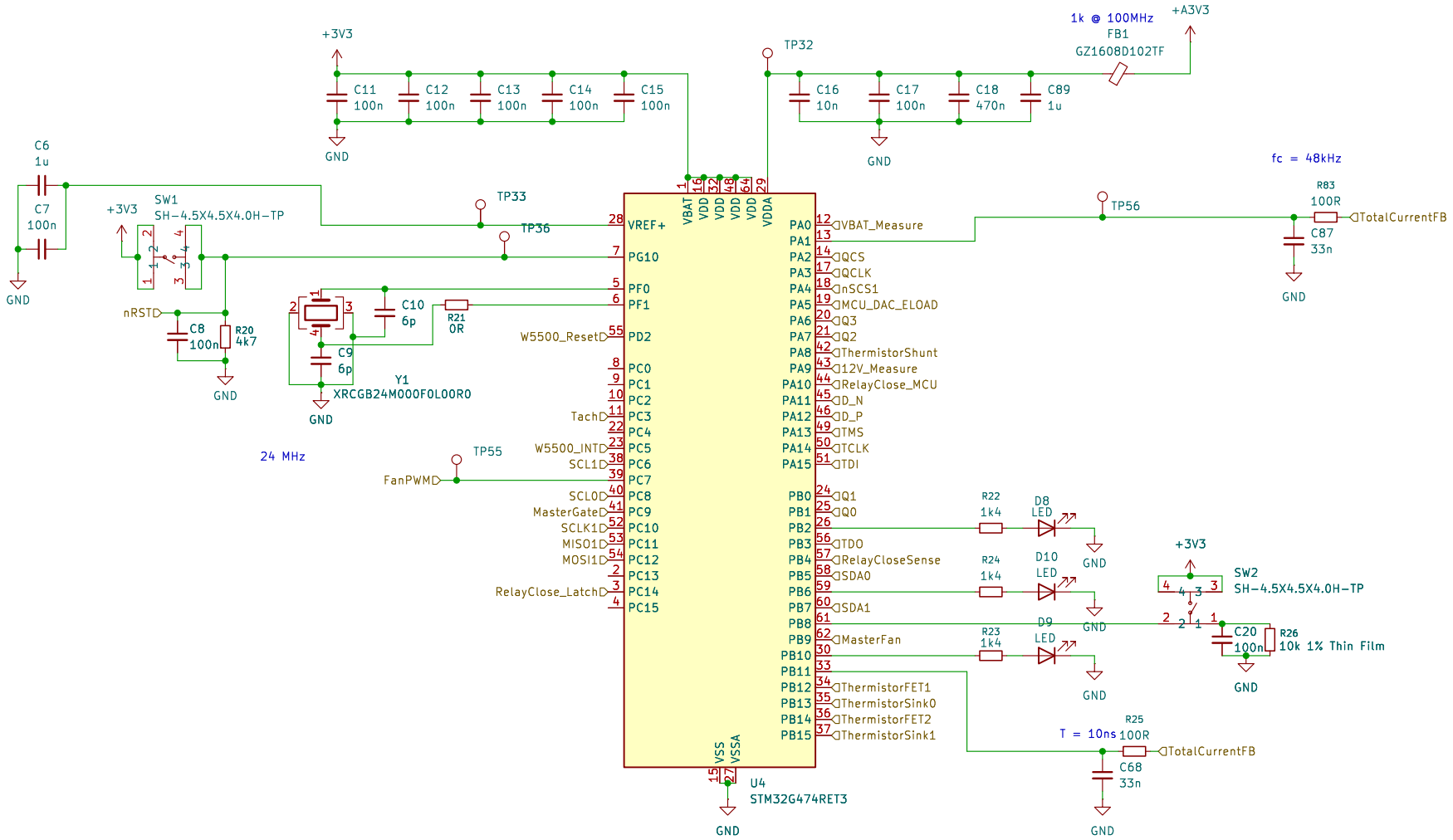




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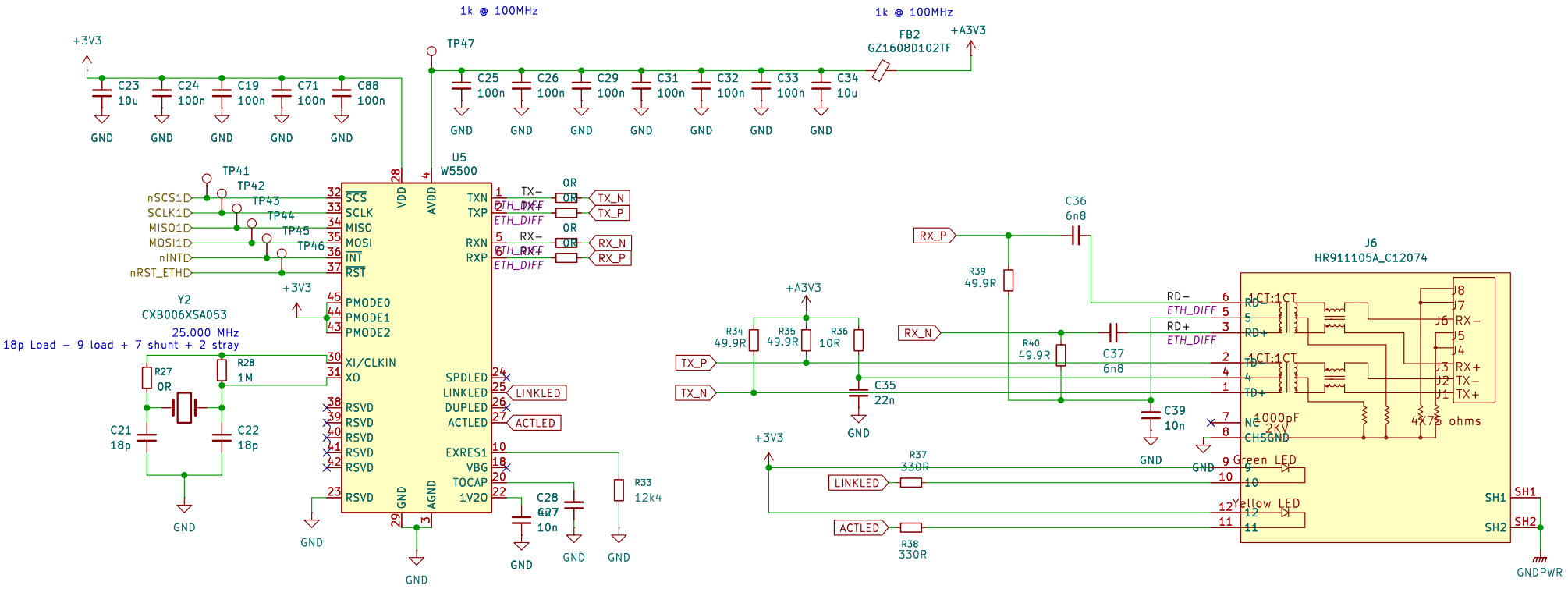
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 14/30

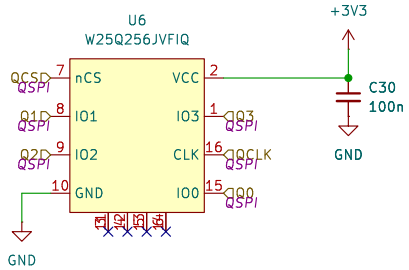


Sheet: /MCU/		
File: MCU.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 5/30

Integrated MAC/PHY – 10/100 BASE-TX

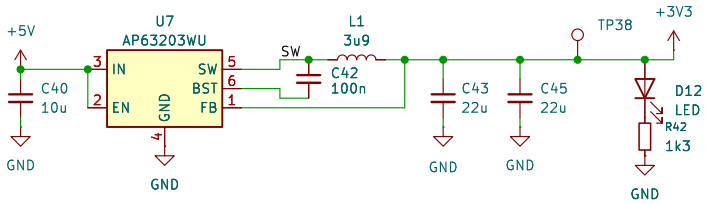


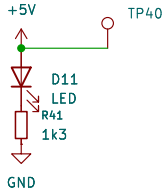
256Mbit NOR Flash



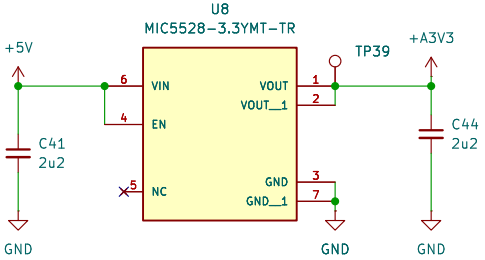
Sheet: /Peripherals/		
File: Peripherals.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 6/30

3.8–32V TO FIXED 3V3 2A MAX, 1.1MHz





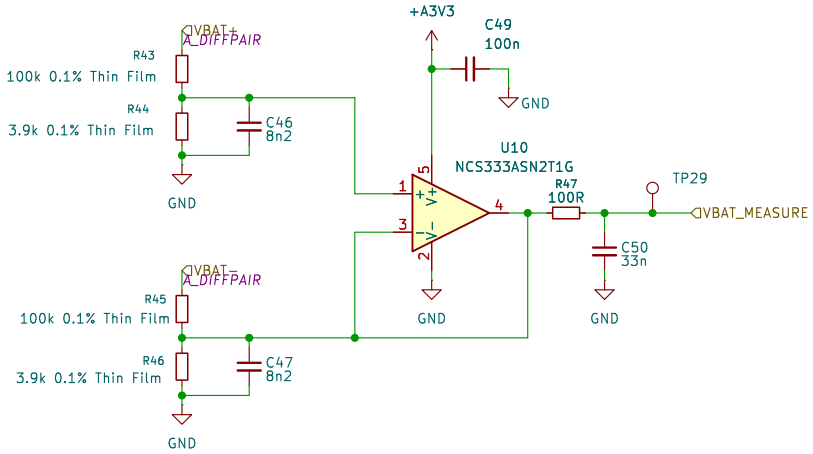
5V TO A3V3 – 500mA MAX



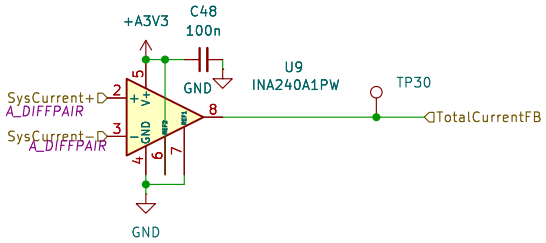
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File: Power.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 7/30

VBAT Voltage Measurement

VBAT MEASUREMENT
R1 = 100k, R2 = 3.9k
GAIN = 0.0375 | A * 3V = 0.11 | A * 85V = 3.19V
INPUT CUTOFF = 5.2kHz | OUTPUT CUTOFF 4.8kHz | INPUT OFFSET 10uV

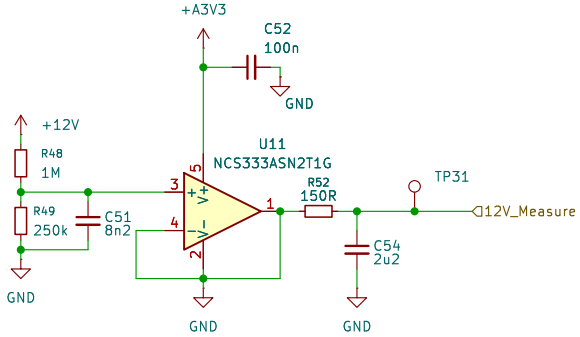


VBAT Current Measurement



12V Rail Measurement

12V SUPPLY MEASUREMENT
R1 = 1M, R2 = 250K
GAIN = 0.2 | A * 16V = 3.2V
INPUT CUTOFF = 100Hz | OUTPUT CUTOFF 482Hz | INPUT OFFSET 10uV



Sheet: /Sensing/		
File: Sensing.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 8/30

EXTERNAL THERMISTOR CHANNELS

ExtThermistor
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor1
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor2
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor7
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor3
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor6
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor4
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor5
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor8
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor11
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor9
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistor10
Thermistor
File: ExtThermistor.kicad_sch

ExtThermistorADC
AIN0
AIN1
AIN2
AIN3
ADDR
SCL1
SDA1
+A3V3
TP52
TP53
SCL1
SDA1
File: ExtThermistorADC.kicad_sch

ExtThermistorADC1
AIN0
AIN1
AIN2
AIN3
ADDR
SCL1
SDA1
GND
File: ExtThermistorADC.kicad_sch

ExtThermistorADC2
AIN0
AIN1
AIN2
AIN3
ADDR
SCL1
SDA1
File: ExtThermistorADC.kicad_sch

NTC Thermistor – High Side Switch 1
B25/50 = 3950K
10k @ 25C
−55C to 150C
Vout: 0.54V @ 20C, 2.93V @ 150C
+A3V3
R53
1k5 1% Thin Film
ThermistorFET1
TH1
MF58 3950B
C55
100n
GND

NTC Thermistor – High Side Switch 2
B25/50 = 3950K
10k @ 25C
−55C to 150C
Vout: 0.54V @ 20C, 2.93V @ 150C
+A3V3
R54
1k5 1% Thin Film
TP54
ThermistorFET2
TH2
MF58 3950B
C56
100n
GND

NTC Thermistor – Shunt
B25/50 = 3950K
10k @ 25C
−55C to 150C
Vout: 0.54V @ 20C, 2.93V @ 150C
+A3V3
R50
1k5 1% Thin Film
ThermistorShunt
TH7
MF58 3950B
C53
100n
GND

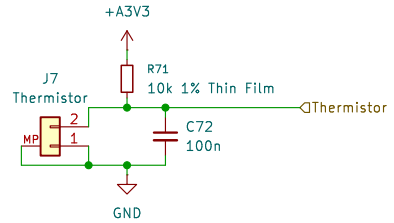
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File: Temperature.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 9/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



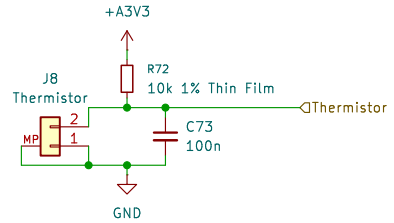
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 15/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



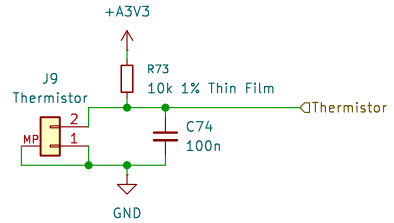
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 16/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



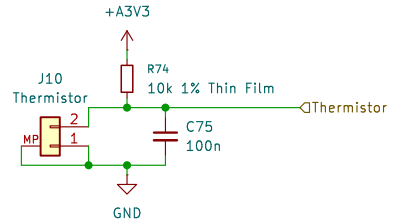
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 17/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



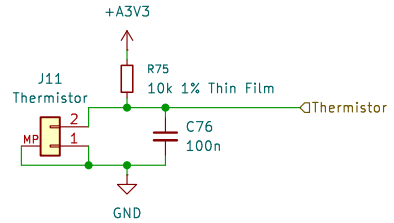
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 18/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



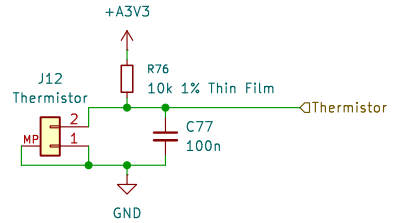
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 19/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



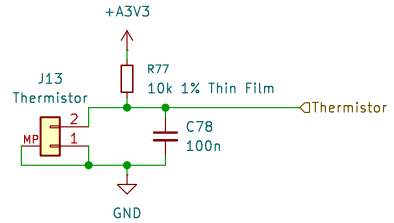
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 20/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



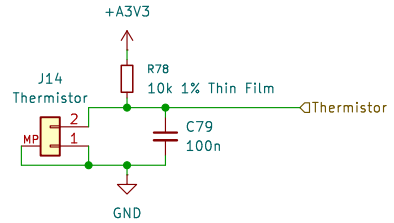
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 21/30

MF52D 103F3950

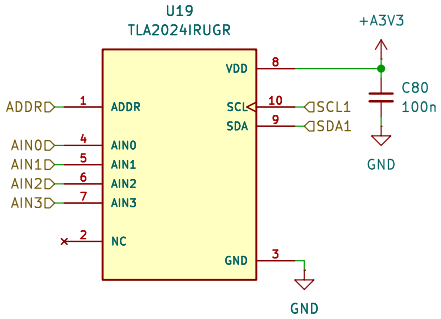
B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



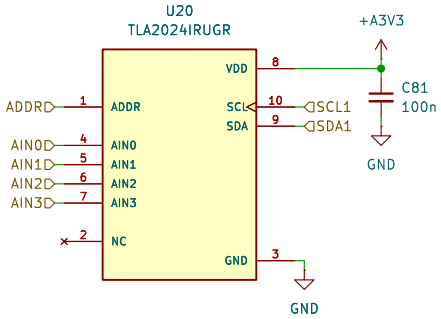
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 22/30



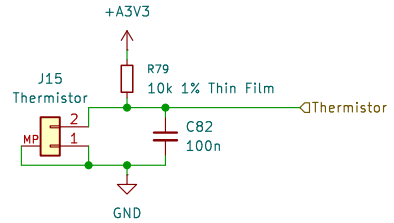
Sheet: /Temperature/ExtThermistorADC/		
File: ExtThermistorADC.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 23/30



Sheet: /Temperature/ExtThermistorADC1/		
File: ExtThermistorADC.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 24/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



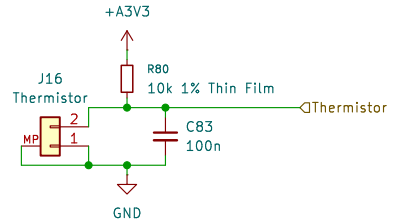
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 25/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



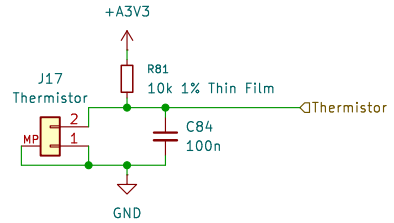
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 26/30

MF52D 103F3950

B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



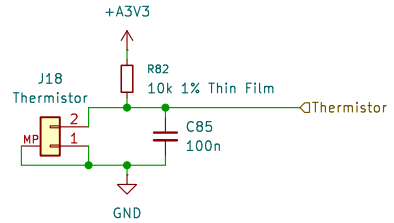
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File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 27/30

MF52D 103F3950

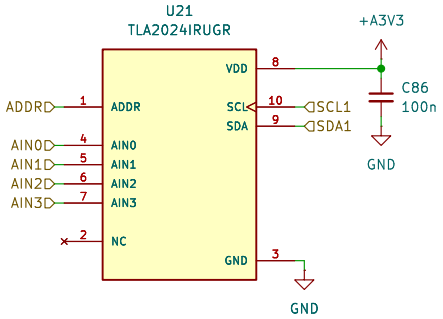
B25/5 = 3950K
10k @ 25C
−30C to 105C
Vout: 1.65V @ 25C, 0.66V @ 60C



Sheet: /Temperature/ExtThermistor11/
File: ExtThermistor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 28/30



Sheet: /Temperature/ExtThermistorADC2/		
File: ExtThermistorADC.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 29/30



Sheet: /Mechanical/
File: Mechanical.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 30/30